

Hours	Frequency
0-1	4,300
1-3	6,900
3-5	4,900
5-10	2,000
10-24	2,100

f
n

Here's the data.

Hours	Frequency	Upper limit	Cumulative frequency
0	0	0	0
0-1	4,300	1	4,300
1-3	6,900	3	$4,300+6,900 = 11,200$
3-5	4,900	5	$4,300+6,900+4,900 = 16,100$
5-10	2,000	10	$4,300+6,900+4,900+2,000 = 18,100$
10-24	2,100	24	$4,300+6,900+4,900+2,000+2,100 = 20,200$

We've added in 0, as you can't play games for LESS than 0 hours a week.

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Drawing the cumulative frequency graph

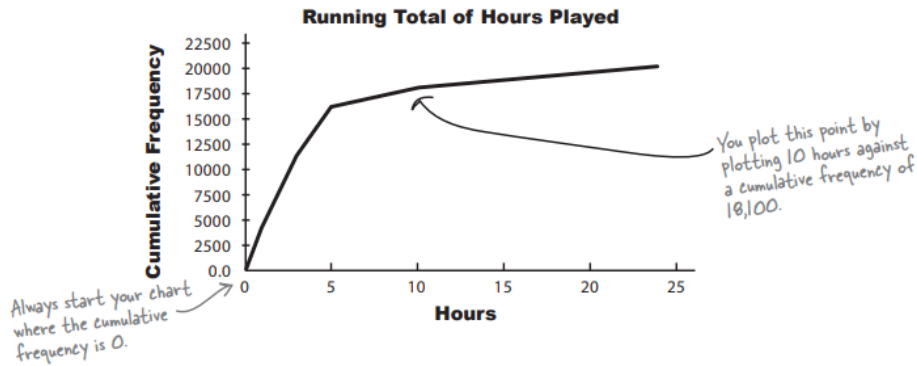
Now that we have the upper limits and cumulative frequencies, we can plot them on a chart. Draw two axes, with the vertical one for the cumulative frequency and the horizontal one for the hours. Once you've done that, plot each of the upper limits against its cumulative frequency, and then join the points together with a line like this:



Watch it!

Cumulative frequencies can never decrease.

If your cumulative frequency decreases at any point, check your calculations.



Q: What's a cumulative frequency?

A: The cumulative frequency of a value is the sum of the frequencies up to and including that value. It tells you the total frequency up to that point.

As an example, suppose you have data telling you how old people are. The cumulative frequency for value 27 tells you how many people there are up to and including age 27.

Q: Are cumulative frequency graphs just for grouped data?

A: Not at all; you can use them for any sort of numeric data. The key thing is whether you want to know the total frequency up to a particular value, or whether you're more interested in the frequencies of particular values instead.

Q: On some charts you can show more than one set of data on the same chart. What about for cumulative frequency graphs?

A: You can do this for cumulative frequency graphs by drawing a separate line for each set of data. If, say, you wanted to compare the cumulative frequencies by gender, you could draw one line showing males and the other females. It would be far more effective to show both lines on one chart, as it makes it easier to compare the two sets of data.

Q: Is there a limit to how many lines you can show on one chart?

A: There's no specific limit, as it all depends on your data. Don't have so many lines that the graph becomes cluttered and you can no longer use it to read off cumulative frequencies and compare sets of data.

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Q: Remind me, how do I find the cumulative frequency of a value?

A: You can find the cumulative frequency by reading it straight off the graph. You locate the value you want to find the cumulative frequency for on the horizontal axis, find where this meets the cumulative frequency curve, and then read the value of cumulative frequency off the vertical axis.

Q: If I already know the cumulative frequency, can I use the graph to find the corresponding value?

A: Yes you can. Look for the cumulative frequency on the vertical axis, find where it meets the cumulative frequency curve, and then read off the value.

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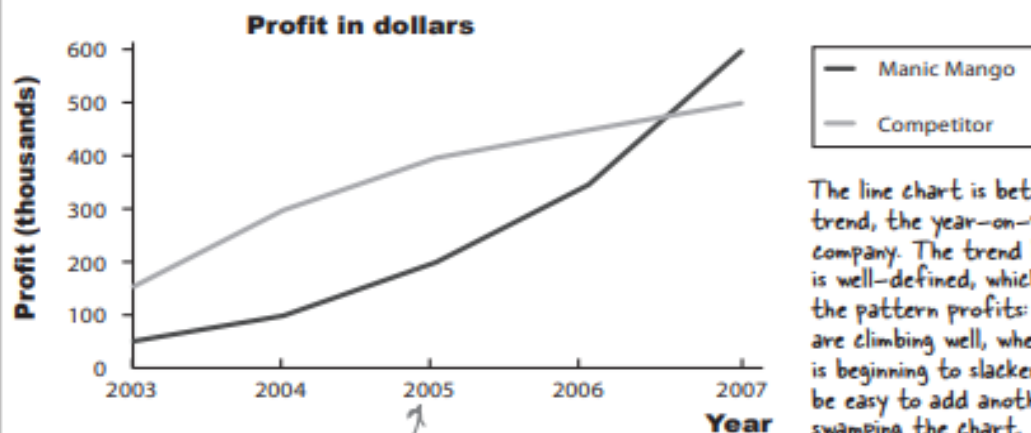


Here are two possible charts that the CEO could use in his keynote. Your task is to annotate each one, and say what you think the strengths and weaknesses are of each one relative to the other. Which would you pick?



The bar chart does a good job of comparing the profit on a year-by-year basis, and it's great if you want to compare profits in an individual year. As an example, we can see that up to 2007, the competitor made a bigger profit, but in 2007 Manic Mango did.

A weakness of this chart is that if the CEO suddenly decided to add a third competitor, it might make the chart a bit harder to take in at a single glance.



The line chart is better at showing a trend, the year-on-year profits for each company. The trend line for each company is well-defined, which means we easily see the pattern profits: Manic Mango profits are climbing well, where its competition is beginning to slacken off. It would also be easy to add another company without swamping the chart.

A weakness is that you can also compare year-by-year profit, but perhaps the bar chart is clearer.

We'd choose the line chart, as the overall trend is clearer than on the bar chart. But don't worry if you chose the other; the chart you use depends on which key facts you want to emphasize.

Line charts are good at showing trends in your data. For each set of data, you plot your points and then join them together with lines. You can easily show multiple sets of data on the same chart without it getting too cluttered. Just make sure it's clear which line is which.

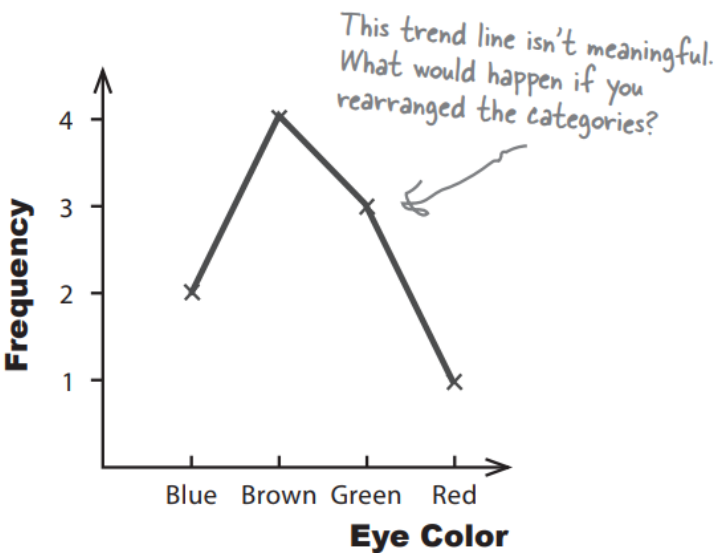
As with other sorts of charts, you have a choice of showing frequency or percentages on the vertical axis. The scale you use all depends on what key facts you want to draw out.

Line charts are often used to show time measurements. Time always goes on the horizontal axis, and frequency on the vertical. **You can read off the frequency for any period of time by choosing the time value on the horizontal axis**, and reading off the corresponding frequency for that point on the line.

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On this arranged data, we may have a different values.

Line charts should be used for numerical data only, and not categorical. This is because it makes sense to compare different categories, but not to draw a trend line. Only use a line chart if you're comparing categories over some numerical unit such as time, and in that case you'd use a separate line for each category.



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BULLET POINTS

- **Cumulative frequency** is the total frequency up to a particular value. It's a running total of the frequencies.
- Use a cumulative frequency graph to plot the upper limit of each group of data against cumulative frequency.
- Use a line chart if you want to show trends, for example over time.
- You can show more than one set of data on a line chart. Use one line for each set of data, and make sure it's clear which line is which.
- You can use line charts to make basic predictions as it's easy to see the shape of the trend. Just extend the trend line, trying to keep the same basic shape.
- **Don't use line charts to show categorical data** unless you're showing trends for each category, for example over time. If you do this, draw one line per category.

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Q: Can line charts show categorical data as well as data that's numeric?

A: Line charts should only be used to show categorical data if you're showing trends for each category, and use a separate line for each category.

What you *shouldn't* do is use a line chart to draw lines from category to category.

Q: So line charts are better for showing overarching trends, and bar charts are better for comparing values or categories?

A: That's right. Which chart you use really comes down to what message you want to put across, and what key facts you want to minimize.

Q: Are line charts the same thing as time series charts? I think I've heard that name used before.

A: A time series chart is really a line chart that focuses on time intervals, just like the examples we used. A line chart doesn't have to focus on just time, though.

Q: Are there any special varieties of line charts?

A: Yes. In fact, you've encountered one of them already. The **cumulative frequency graph** is a type of line chart that shows the total frequency up to a certain value

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One way to find **the average** of a bunch of numbers is **to add** all the numbers together and then **divide by** how many numbers there are. In statistics, this is called the **mean**.

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Letters and numbers

Almost every statistical calculation involves adding a bunch of numbers together. As an example, if we want to find the mean of the Power Workout class, we first have to add the ages of all the class attendees together.

The problem statisticians have is how to generalize this. We don't necessarily know in advance how many numbers we're dealing with, or what they are. We currently know how many people are in the Power Workout class and what their ages are, but what if someone else joins the class? If we could only generalize this, we'd have a way of showing the calculation without rewriting it every time the class changes.

Statisticians get around this problem by using *letters* to represent *numbers*. As an example, they might use the letter **x** to represent ages in the Power Workout class like this:

Specific ages of class attendees

19 20 20 20 21

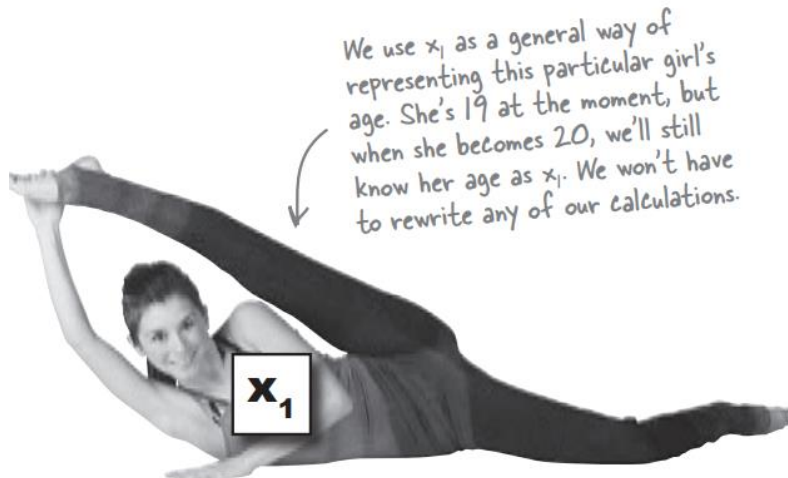


General ages of class attendees

x_1 x_2 x_3 x_4 x_5

Each **x** represents the age of a separate person in the class. It's a bit like labeling each person with a particular number **x**.

Each **x** represents one of the class ages.



Now that we have a general way of writing ages, we can use our **x**'s to represent them in calculations. We can write the sum of the 5 ages in the class as

$$\text{Sum} = x_1 + x_2 + x_3 + x_4 + x_5$$

But what if we don't know how many numbers we have to sum? What if we don't know how many people are in the class?

Dealing with unknowns

Statisticians use letters to represent unknown numbers. But what if we don't know how many numbers we might have to add together? Not a problem—we'll just call the number of values n . If we didn't know how many people were in the Power Workout class, we'd just say that there were n of them, and write the sum of all the ages as:

$$\text{Sum} = x_1 + x_2 + x_3 + x_4 + x_5 + \dots + x_n$$

The " $\dots$ " is a quick way of saying "and so on." In other words, just keep on adding x 's.

In this case, x_n represents the age of the n th person in the class. If there were 18 people in the class, this would be x_{18} , the age of the 18th person.

We can take another shortcut.

Writing $x_1 + x_2 + x_3 + x_4 + \dots + x_n$ is a bit like saying "add age 1 to age 2, then add age 3, then add age 4, and keep on adding ages up to age n ." In day-to-day conversation it's unlikely we'd phrase it like this. We're far more likely to say "add together all of the ages." It's quicker, simpler, and to the point.

We can do something similar in math notation by using the summation symbol Σ , which is the Greek letter Sigma. We can use Σx (pronounced "sigma x") as a quick way of saying "add together the values of all the x 's."

It all adds up now...

$$x_1 + x_2 + x_3 + x_4 + x_5 + \dots + x_n = \Sigma x$$

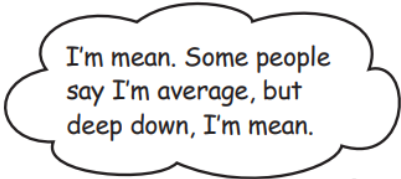
$$\frac{\Sigma x}{n}$$

← Add all the numbers together...

← ...then divide by however many there are.

The mean has its own symbol

The mean is one of the most commonly used statistics around, and statisticians use it so frequently that they've given it a symbol all of its own: μ . This is the Greek letter mu (pronounced "mew"). Remember, it's just a quick way of representing the mean.



I'm mean. Some people say I'm average, but deep down, I'm mean.

$$\mu = \frac{\sum x}{n}$$

The mean is one of the most frequently used statistics. It can be represented with the symbol μ .

Handling frequencies

When you calculate the mean of a set of numbers, you'll often find that some of the numbers are repeated. If you look at the ages of the Power Workout class, you'll see we actually have 3 people of age 20.

It's really important to make sure that you include the *frequency* of each number when you're working out the mean. To make sure we don't overlook it, we can include it in our formula.

If we use the letter **f** to represent frequency, we can rewrite the mean as

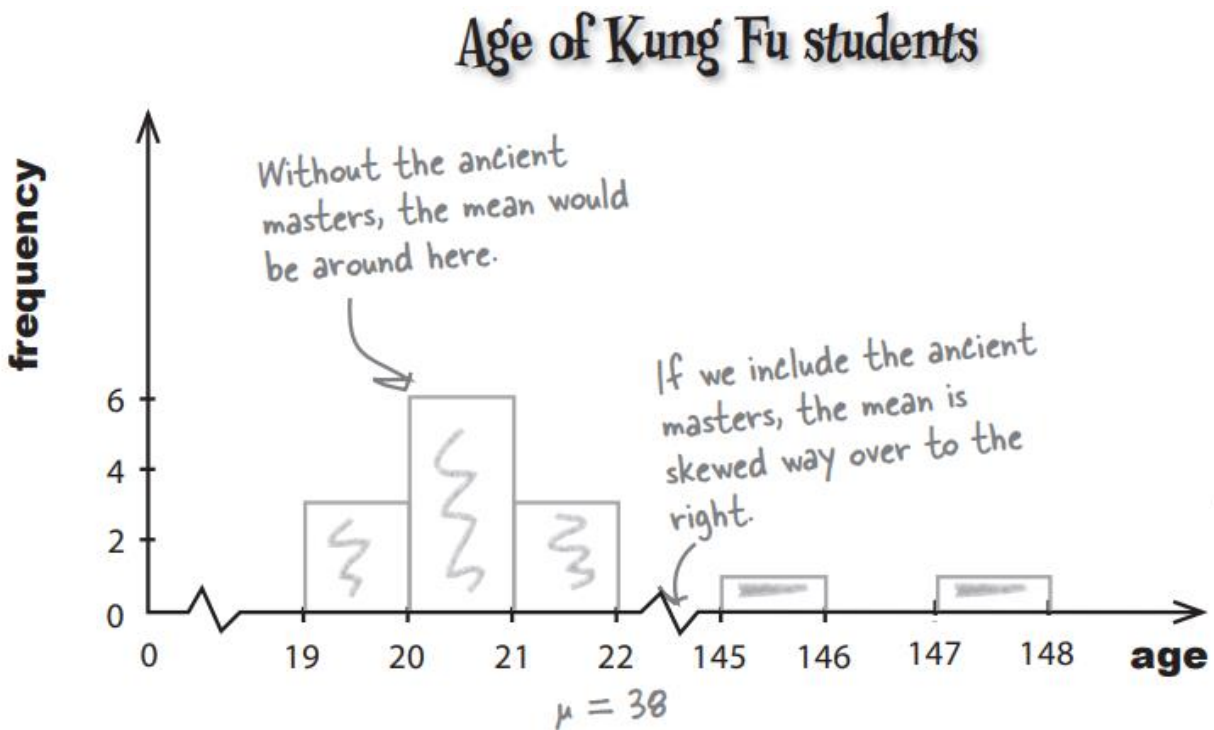
$$\mu = \frac{\sum fx}{\sum f}$$

Multiply each number by its frequency, then add the results together.

Sum of the frequencies

Outlier

An extreme high or low value
that stands out from the
rest of the data



Skewed Data

When outliers "pull" the data
to the left or right